



Western University Faculty of Engineering

Lab Section 20

Team Identifier: 20 – Sound

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Design Documentation: Wildlife Soundscape Monitoring System

A low-cost, deployable outdoor device that captures acoustic data in WAV format to support biodiversity conservation and ecosystem health assessment on the Western University campus.

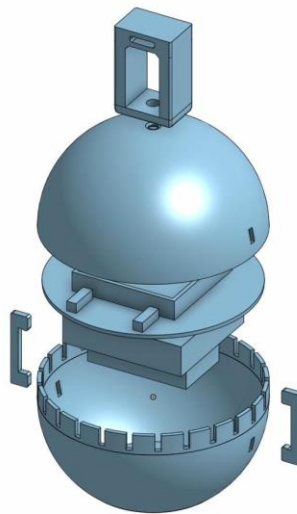


Figure 1: Assembled Wildlife Soundscape Monitoring Device

Final Design Documentation

Need Statement

“Behavioural ecologists and related researchers need a continuous and reliable method to measure noise in urban natural spaces because there is currently no system capable of capturing its impacts on animal behaviour and biodiversity.”

Urban development causes habitat destruction and degradation for wildlife, with noise being one of the primary mechanisms by which habitat quality decreases around urban environments.

User and Insight

The primary user is a Western University behavioural ecology professor and their students, who study how anthropogenic noise affects animal behaviour in urban natural spaces. The professor’s research focuses on local species (cardinals, grey squirrels, house sparrows) and requires acoustic recordings that can be analyzed for frequency content and amplitude over extended deployment periods. The key user insight is that low maintenance targeting over one month between service visits is the single greatest gain over existing methods, followed by data quality and, secondarily, cost.

Why Current Solutions Are Insufficient

The current commercial alternatives are the Wildlife Acoustics Micro Recorder (~\$200/unit) and the AudioMoth (\$125/unit). Both require frequent manual data retrieval (approximately every two weeks), suffer from unreliable Bluetooth connectivity, have challenging app interfaces, and do not measure amplitude reliably. Neither offers true 360-degree omnidirectional audio capture. The AudioMoth in particular lacks amplitude measurement a critical data point for the client’s research. Our system targets all of these gaps at a comparable or lower unit cost (~\$77.50).

Key Design Criteria

Key constraints include: frequency capture from 60–20,000 Hz; minimum detectable sound level of 40 dB; operating temperature of -20°C to 30°C; omnidirectional audio capture; WAV file output; MicroSD card storage; weather-resistant housing; deployable by a single person; under 5 lbs excluding mounting hardware; unit cost under \$200; and must record data in chunks of 5 minutes every hour. Key objectives include maximizing degrees of sound capture, maximizing portability, maximizing reliability, maximizing run time storage, maximizing battery capacity, minimizing maintenance, minimizing repair complexity, minimizing cost, minimizing power consumption, minimizing environmental impact, maximizing durability, and maximizing ease of use.

Description

The Wildlife Soundscape Monitoring System is designed to continuously capture ambient audio data in outdoor environments for behavioral ecology research. The system records 5-minute WAV audio clips on a scheduled basis, storing them on a removable microSD card for later retrieval and analysis using software such as Cornell’s Raven Pro.

The device features a fully spherical 3D-printed enclosure that enables near-360-degree sound capture through an equatorial mesh opening. A modular tether connection system at the top allows flexible mounting via a hose clamp and adjustable fishing wire, enabling deployment on trees, poles, or similar structures at 5–7 feet above ground. All internal hardware is organized into two

layers: a sealed electronics compartment (ESP32 microcontroller, ICS-43434 MEMS microphone, MicroSD adapter, DC-DC buck converter) and a user-accessible compartment for tool-free battery and microSD card swapping. The two halves of the outer casing are fastened together using a clip mechanism for easy field access.

Design Drawings

Overall Schematic Drawing

The system architecture consists of the following signal and power chain: AA batteries feed a DC-DC buck converter (LM2596) which regulates voltage to 3.3V for the ESP32 WROOM microcontroller. The ESP32 reads digital audio data from the ICS-43434 MEMS microphone via the I2S protocol, formats it with a WAV file header, and writes the resulting file to a MicroSD card through an SPI-connected card adapter. The equatorial mesh barrier on the spherical housing allows omnidirectional sound to reach the microphone while protecting against debris ingress.

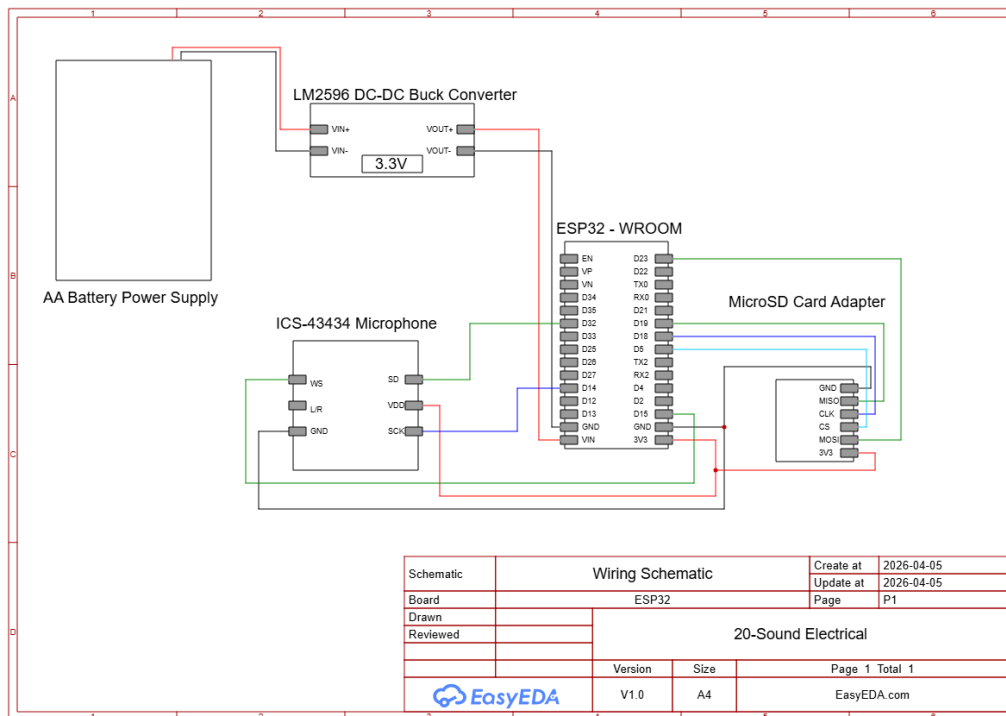


Figure 2: Wiring Schematic — ESP32, ICS-43434 Microphone, MicroSD Adapter, and LM2596 DC-DC Buck Converter

Overall Layout Drawing

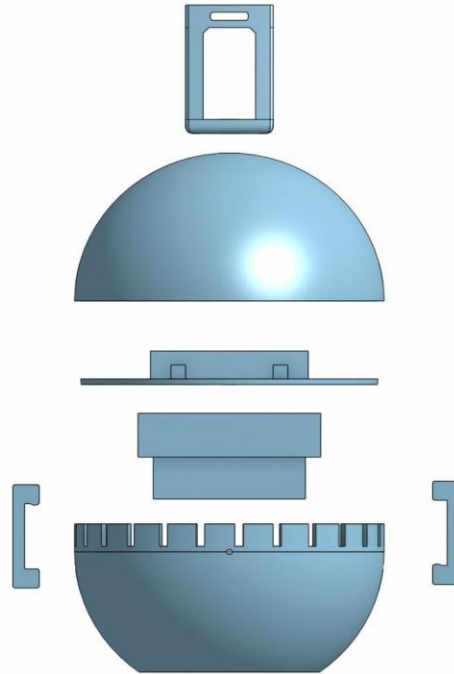


Figure 3: Exploded View of Sphere Assembly

The exploded view above shows the spatial arrangement of all physical components. From top to bottom: the modular tether connection bracket (hose clamp insert with bolt and string insert), the top hemisphere shell, the microphone plate (which seats the electronics), the middle mesh barrier ring for sound penetration, the electrical storage plate (electronics mounting surface), and the bottom hemisphere shell with integrated clip arm slots. Two clip arms on the lower hemisphere secure the halves together.

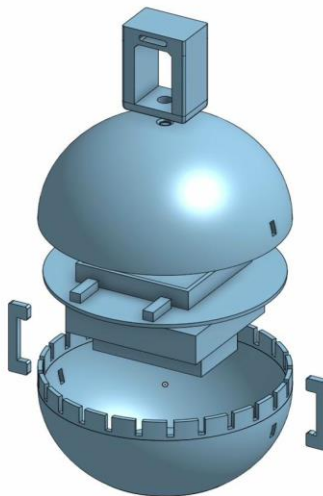


Figure 4: Assembled Isometric View

Overall Process Diagram

The firmware process follows this sequence: (1) Boot and initialize the ESP32, setting up the microphone via I2S and microSD card module via SPI. (2) Create a session folder named "AUDIO_LOGGER_SESSION_<number>" on the microSD card to store all files. (3) Enter the main loop with a 60-second cycle: if the microSD card is writable, open a new sequentially numbered .wav file, encode the raw I2S data, and close the file; otherwise delay briefly. (4) Enter sleep mode for the remaining time in the cycle, then return to step (3) and repeat.

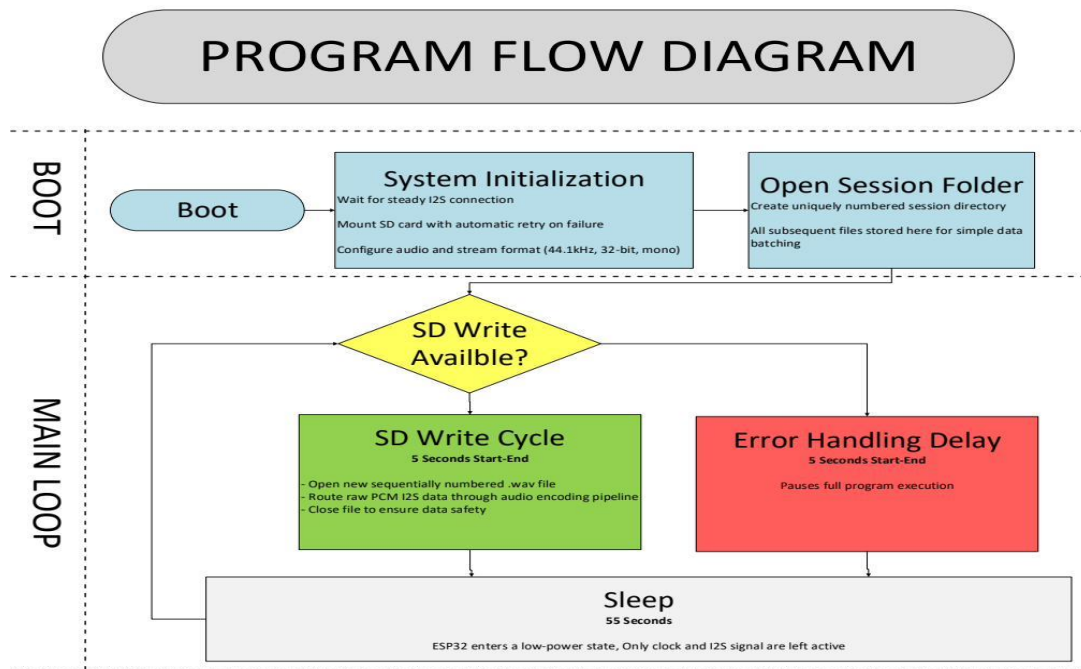


Figure 5: Program Flow Diagram — Boot Sequence and Main Recording Loop

Note: In the current prototype, the loop runs on a 60-second time, mimicking the 60-minute cycle time of our final design, yet enabling extended demo and testing runs of 50+ recordings.

Detail Drawings

Part Drawings

All parts were designed in Onshape and 3D-printed in PLA filament. The following table summarizes the individual printed components:

#	Part	Mass	Price	Material
1	Hose Clamp Mount	18.36 g	\$2.75	PLA
2	Top Hemisphere	42.32 g	\$6.35	PLA
3	Middle Mesh Barrier	6.46 g	\$0.97	PLA
4	Bottom Hemisphere	41.59 g	\$6.24	PLA
5	Bottom Sphere Electronic Plate	32.45 g	\$4.87	PLA
6	Top Sphere Plate	10.35 g	\$1.55	PLA
7	Clip Arms (×2)	0.81 g	\$0.24	PLA
	Total	152.34 g	\$22.97	

Table 1: 3D-Printed Part Summary

Assembly Drawings

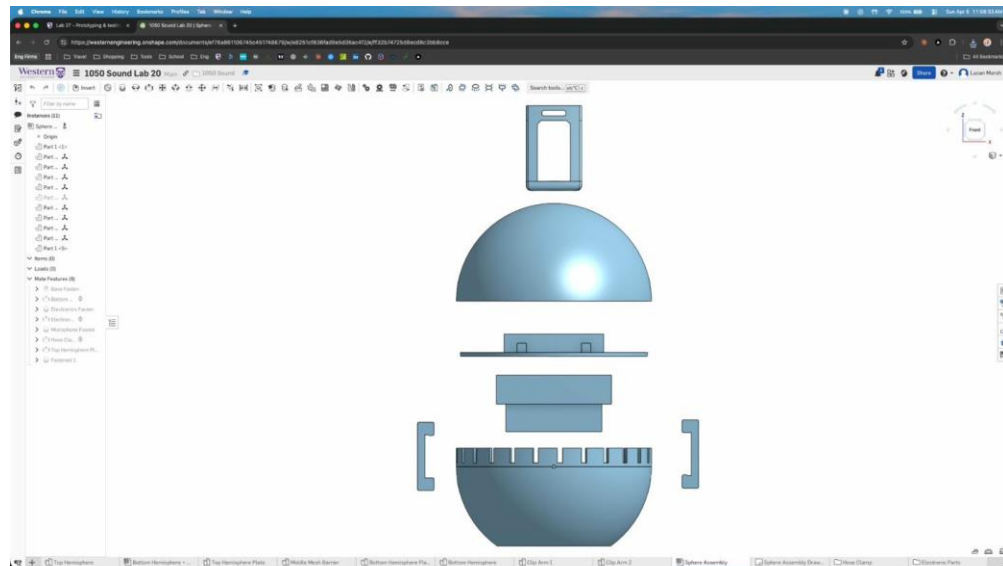


Figure 6: Onshape Assembly Environment with Mate Features

The assembly consists of 11 instances with 8 mate features including: Base Fastener, Bottom Hemisphere Plate fastener, Electronics Fastener, Microphone Fastener, Hose Clamp fastener, and Top Hemisphere Plate fastener.

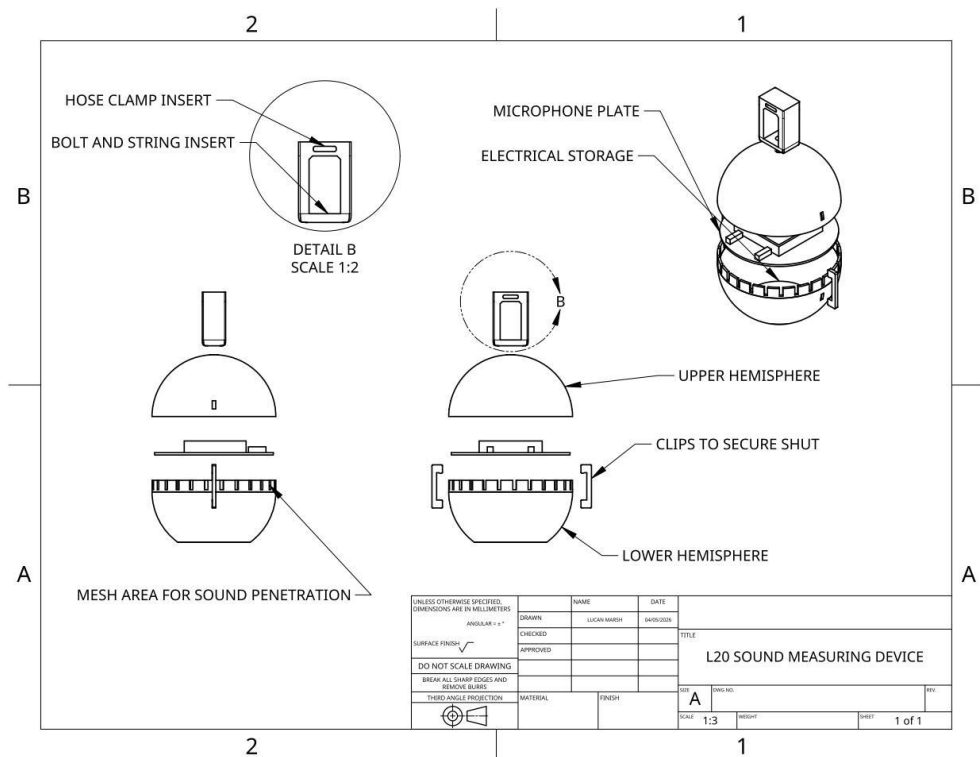


Figure 7: Annotated Assembly Drawing — Part Callouts and Key Features

The annotated assembly drawing identifies the major subassemblies: the hose clamp insert and bolt/string insert at the top, the microphone plate and electrical storage compartment in the centre, the clips that secure the two halves shut, the upper and lower hemisphere shells, and the mesh area along the equator for sound penetration.

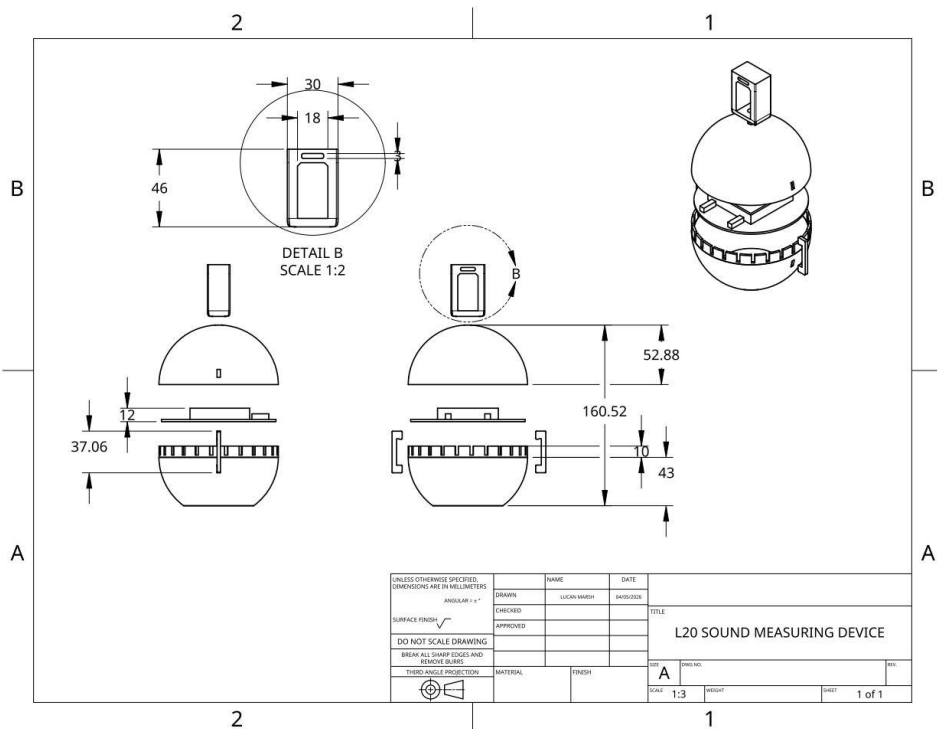


Figure 8: Dimensioned Assembly Drawing — Overall Dimensions and Detail Views

The dimensioned assembly drawing shows the overall device height of approximately 160.52 mm, a hemisphere diameter of approximately 52.88 mm, and a tether bracket width of 30 mm. Detail B at 1:2 scale shows the tether connection bracket geometry. All dimensions are in millimetres.

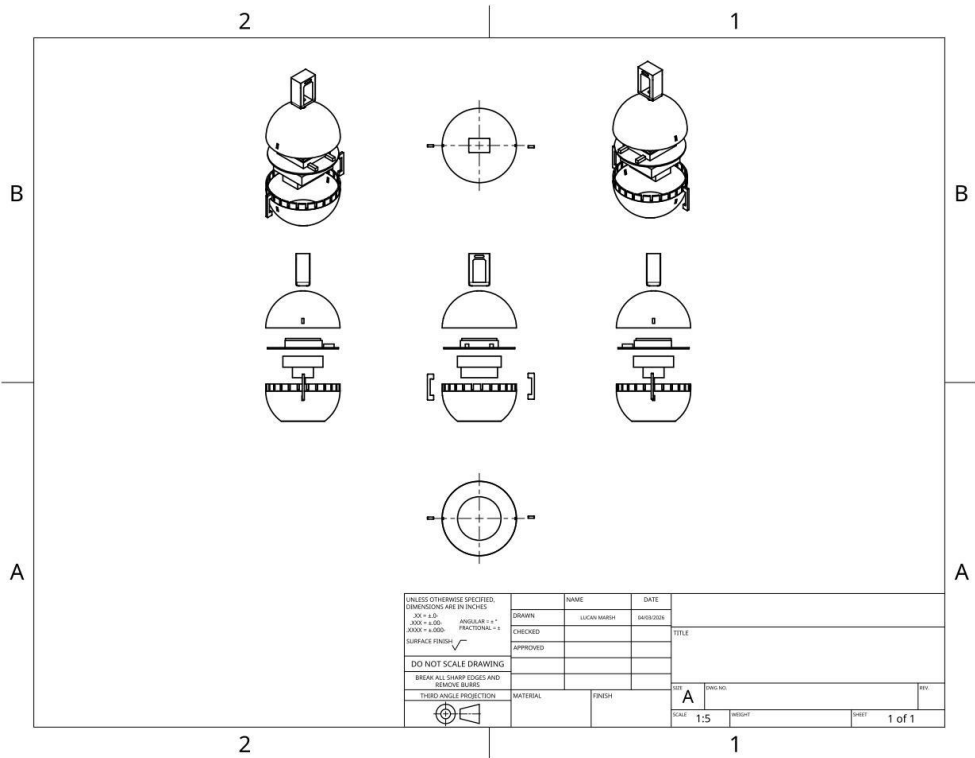


Figure 9: Multi-View Assembly Drawing — Orthographic Projections at 1:5 Scale

Software or Microelectronic Device Code

Code Overview

The firmware is developed for the ESP32 WROOM microcontroller using the Arduino IDE framework. The code is organized into the following functional modules:

- **I2S Driver Configuration:** Initializes the I2S peripheral to interface with the ICS-43434 MEMS microphone, configuring sample rate (44.1 kHz), bit depth (32-bit), and stereo channel format.
- **WAV Header Generator:** Constructs a standard 44-byte WAV file header with the appropriate sample rate, bit depth, and data size parameters.
- **MicroSD Card Writer:** Manages SPI communication with the MicroSD card adapter, handling file creation, sequential numbering, data writing, and safe file closure. Includes automatic retry on mount failure.
- **Sleep/Wake Controller:** Manages 55-minute low-power sleep intervals between 5-minute recording cycles, keeping only the clock and I2S signal active during sleep.
- **Session Manager:** Creates uniquely numbered session directories on the microSD card for organized data batching across recording sessions.

Repository & Setup

Repository link: https://github.com/NEhnes/mic_sd_logger.

License: MIT 2.0 (allowing for free distribution, modification, etc.)

Dependencies:

- Arduino-Audio-Tools v1.2.2 (audio handling)
- Platformio (build system)
- SD (built-in arduino library)
- SPI (built-in arduino library)

Conventions: camelCase variables,

The repository contains more detailed setup instructions, however, here is a simple script that can be pasted into a terminal to download all necessary toolchains & upload to a connected ESP32.

```
# install platformio toolchain
sudo apt install platformio

# install git
sudo apt install git

# clone repository locally
git clone https://github.com/NEhnes/mic_sd_logger

# navigate to repository
cd mic_sd_logger

# build & upload to connected ESP32
pio run -t upload
```

Bill of Materials

The following table presents the Solution Bill of Materials for the final design concept (not the prototype). The total unit cost of \$77.50 is well within the \$200 per unit budget constraint.

#	Item Description	Units	Qty	Unit Cost	Cost	Source	
1	ICS-43434 MEMS Digital Microphone Module	1	1	\$11.63	\$11.63	Amazon	
2	ESP32 WROOM MCU	1	1	\$13.99	\$13.99	Amazon	
3	MicroSD Card Adapter	1	1	\$2.00	\$2.00	In-House	
4	DC-DC Buck Converter LM2596	1	1	\$6.00	\$6.00	In-House	
5	AA Battery Holder	1	2	\$4.50	\$9.00	Amazon	
6	Mesh	1	1	\$11.69	\$11.69	Amazon	
7	Braided Fishing Line	1	1	\$8.49	\$8.49	Amazon	
8	Hose Clamp	1	1	\$10.28	\$10.28	Amazon	
9	1/4 Inch Bolt	1	1	\$0.57	\$0.57	In-House	
10	1/4 Inch Nut	1	1	\$0.85	\$0.85	In-House	
11	22 AWG Solid Wire	1	1	\$3.00	\$3.00	In-House	
				Total:	\$77.50		

Table 2: Solution Bill of Materials

Note: 3D printing costs (\$22.97 total for all printed parts) and filament costs (\$6.00) are accounted for separately in the Prototype Bill of Materials. The solution BOM reflects the components required for a single production unit.

User Instructions (Operations and Maintenance)

Setup and Deployment

1. Insert a formatted MicroSD card (FAT32, 32–64 GB recommended) into the card slot in the user-accessible compartment.
2. Insert fresh AA batteries into the battery holders, ensuring correct polarity.
3. Close the outer casing by aligning the two hemisphere halves and engaging the clip arms on both sides until they click into place.
4. Select the deployment location: a tree, pole, or structure 5–7 feet above ground, away from large reflective surfaces to minimize echo.
5. Secure the hose clamp around the chosen structure and tighten. Attach the mounting block to the hose clamp.
6. Thread the braided fishing line from the mounting block through the tether bracket on top of the device. Adjust the line length, so the device hangs freely at the desired height.
7. The device will begin recording automatically according to its programmed schedule.

Data Retrieval

8. Carefully unclip the outer casing by releasing both clip arms.
9. Remove the MicroSD card from the user-accessible compartment.
10. Insert a replacement MicroSD card if continued recording is desired.
11. Close the casing and re-engage the clips.
12. Transfer WAV files from the retrieved MicroSD card to a computer for analysis in Raven Pro, Excel, or similar software.

Maintenance

Battery replacement: Replace AA batteries when the device stops recording or at scheduled intervals (estimated every 4–8 weeks depending on the recording schedule and ambient temperature).

MicroSD card management: Swap MicroSD cards before they reach capacity. A 32 GB card holds approximately 50+ hours of uncompressed stereo WAV audio at 132-bit, 44.1 kHz.

Enclosure inspection: Check the mesh barrier for debris accumulation and clean as needed. Inspect the clip arms and tether connection for wear or damage.

Desiccant replacement: Replace silica gel desiccant packets inside the sealed electronics compartment every 2–3 months or after heavy rain/humidity exposure.

Troubleshooting

Device not recording: Confirm battery polarity and charge level. Verify the microSD card is properly seated and formatted as FAT32. Check that the card is not write-protected.

Audio files are corrupted: Ensure the device is not powered off or opened during an active recording. Replace the microSD card if corruption persists.

Poor audio quality: Inspect the mesh barrier for blockage. Ensure the device is hanging freely and not pressed against a surface.

Testing

The following tables document software and physical tests conducted to validate the design.

Software Tests

#	Test Name	Purpose	Outcomes	Insights / Lessons
1	Beastbroak repo test	Evaluate existing open-source I2S audio code	Code did not compile; no serial communication	Too large/complex as starting point; abandoned.
2	Atomic repo test	Evaluate simpler I2S codebase for ESP32	Serial communication achieved on Wokwi	Much more manageable; selected as base.
3	Hardware integration test	Verify physical ESP32 + INMP441 wiring	Initial wiring confirmed	Hardware connections validated.
4	Verify dummy I2S data	Confirm temporary I2S dummy data source	All metrics passed; repeatable results	Sine wave generator validated downstream pipeline.
5	I2S to Serial pipeline	Ensure reliable I2S frame reading	Serial output received; desired output confirmed	Copier class versatile for SD file writing.

Table 3: Software Testing Results

Physical and Functional Tests

#	Test Name	Description	Result	Insights / Lessons
1	Portability test	Verify device can be carried in one hand	Passed (27.71s deploy time)	Lightweight and compact for single-hand carry.
2	Durability / drop test	Drop test at 15", 30", and 45"	Clip broke after 45" drop	Clip design weak point; needs redesign.
3	Weight test	Verify under 5 lbs (scale measurement)	Passed — under 5 lbs	Weight constraint met.
4	Deploy ability test	Verify single-person deployment	Passed	Single-person deployment confirmed.
5	Maintainability test	Verify single-person maintenance	Passed	Tool-free access confirmed.
6	Weather resistance	Hair dryer on hot/cold settings (~30s)	Passed — no differences	Extreme temp testing (-20°C/30°C) still needed.
7	Reliability test	Extended deployment test	Ran ~600s in test	Longer tests needed for field deployment.

Table 4: Physical and Functional Testing Results

Sustainability, Design Principles, and Lessons Learned

UN Sustainable Development Goals

SDG 15 – Life on Land: The device directly supports biodiversity monitoring by providing researchers with acoustic data to study how urban noise impacts wildlife behavior and habitat quality. Long-term deployment generates the longitudinal datasets needed to detect ecosystem health changes over time.

SDG 13 – Climate Action: By enabling multi-site, long-term acoustic monitoring at low cost, the system supports research into how climate-related changes to urban soundscapes affect animal behavior. The design minimizes environmental impact through a low-power duty cycle, recyclable 3D-printed PLA materials, and a modular architecture that reduces electronic waste through component-level repairability.

SDG 9 – Industry, Innovation and Infrastructure: The project demonstrates how accessible, low-cost open hardware (ESP32, MEMS microphones, open firmware) can be combined into a purpose-built scientific instrument that outperforms or matches commercial alternatives costing two to four times as much, making acoustic monitoring infrastructure more attainable for under-resourced research programs.

Engineering and Universal Design Principles

The design integrates several good engineering design principles. Modularity is embedded throughout: the two-hemisphere clip system, the separable electronics and battery compartments, and the hose clamp tether each allow independent replacement without disturbing other subsystems. The duty-cycle architecture dramatically reduces power draw and storage consumption, directly addressing the client's primary concern of maintenance frequency. The spherical enclosure geometry eliminates flat horizontal surfaces that accumulate snow and debris, improving year-round reliability without added complexity.

Universal design principles are reflected in the tool-free battery and microSD card access — the user-accessible compartment allows a single person to service the device without specialized tools or training. The adjustable hose clamp and fishing line tether system accommodates a wide range of mounting structures and heights. The neutral, non-threatening spherical form factor avoids resembling food sources or attracting animal interference while remaining visually identifiable to humans as a research device.

Risks

The following table presents the top 10 risks by Risk Priority Number (RPN = Likelihood × Severity × Detectability, each scored 1–10).

#	Risk / Failure Mode	L	S	D	RPN	Action	Rationale / Plan
1	Condensation on electronics	6	9	9	486	Reduce	Silica gel desiccant packets inside sealed enclosure
2	Firmware failure	6	8	9	432	Avoid	Team code reviews and edge-case testing
3	Battery rupture due to temperature	6	7	8	336	Avoid	Use appropriate battery chemistry for extreme temps
4	Brown-out	7	6	8	336	Avoid	Add capacitors for power supply stability
5	Microphone out of calibration	4	9	9	324	Accept	Scheduled calibration intervals

6	Static discharge wrecks components	5	9	7	315	Avoid	Non-conducting shell material
7	Mechanical stress on microSD slot	8	7	5	280	Reduce	Design internals for ergonomic microSD access
8	Batteries inserted backwards	9	9	3	243	Accept	Circuit fail-safes such as protection diodes
9	MicroSD card failure (corruption)	5	8	6	240	Accept	Rewrite/replace as needed
10	Vandalism	3	9	8	216	Reduce	Camouflage colouring and elevated mounting

Table 5: Top 10 Risks by RPN

Lessons Learned

The firmware development process surfaced with several key lessons. Beginning with a large, feature-rich open-source I2S codebase (Beastbroak) proved counterproductive — compile errors and unclear architecture made it difficult to isolate problems. Switching to a simpler, minimal repository (Atomic) allowed the team to achieve serial communication rapidly and build up from a working baseline. This reinforced the engineering principle of starting with the simplest system that demonstrates core functionality before adding complexity.

On the hardware side, using a sine wave dummy data generator to validate the I2S-to-SD pipeline before connecting live microphone hardware significantly accelerated testing. This decoupling of subsystem validation from full system integration is a best practice the team will carry forward.

On the physical design side, the initial enclosure exceeded the target hand-sized footprint, highlighting the need to establish dimensional constraints earlier in the iterative CAD process. The drop test revealed that the clip arm design is a weak point — one clip broke after a 45-inch drop. While the clips are a temporary prototype feature, this informed the need for a more robust latching mechanism in the final production design.

Validating Compliance

Constraints Validation

#	Constraint	Assessment	Rationale / Evidence or Plan
1	Must measure under 10 kHz	Not verified yet	ICS-43434 claims response up to 20 kHz; requires field verification.
2	Must survive -20°C to 30°C	Not met	Hair dryer test showed no differences; extreme cold testing was not conducted.
3	Must be under \$200 per unit	Met	Solution BOM total is \$77.50, well under the \$200 ceiling.
4	Must not be down for >1 day	Not verified yet	Designed for quick swap but not field tested for extended uptime.
5	Must be between 5–7 ft	Met	Hose clamp mounting allows adjustable height placement.
6	Must be hand sized (8×8×18 cm)	Not met	Current casing exceeds target envelope; size reduction possible.
7	Must be under 5 lbs	Met	Device weighed on scale and confirmed under 5 lbs.
8	Data compatible with Excel	Not verified yet	WAV files can be post-processed into .xlsx/.csv; needs testing.
9	Must not record under 40 dB	Not verified yet	Threshold gating has not yet been implemented in firmware.
10	Record 5 min chunks every hour	Not met	Scheduled recording logic not implemented; uses 5-second cycles.
11	Deployable by 1 person	Met	Deployment tests confirmed single-person installation.
12	Maintainable by 1 person	Met	Maintenance tests confirmed single-person battery/microSD swap.

Table 6: Constraints Compliance Matrix

Objectives Validation

#	Objective	Assessment	Rationale / Evidence or Plan
1	Maximize degrees of sound	Partially met	Maximum 360° openings modelled; near-omnidirectional capture achieved.
2	Maximize portability	Met	Can be carried in one hand (portability test: 27.71 s).
3	Maximize reliability	Met	Extended deployment test (~600s) verified operation.
4	Maximize run time storage	Not verified yet	32 GB card provides months of recordings at current duty cycle.
5	Maximize battery capacity	Not verified yet	AA holders used; higher-capacity cells possible.
6	Minimize looking like food	Met	Spherical design with neutral coloring; not food-like.
7	Minimize maintenance	Partially met	Tool-free swap tested; scheduled recording firmware still needed.
8	Minimize repair	Partially met	Modular internals allow component replacement.
9	Minimize cost	Met	\$77.50/unit, well under \$200 target.
10	Minimize power consumption	Not verified yet	Sleep firmware implemented; ESP32 supports ~10 µA sleep.

11	Minimize environmental impact	Not verified yet	PLA shell is recyclable; battery waste is primary concern.
12	Maximize durability	Partially met	Drop test at 45" caused clip breakage; redesign needed.
13	Maximize ease of use	Met	One person deployed without prior instructions.

Table 7: Objectives Compliance Matrix

Appendix A – Assembly Drawing References

Detailed dimensioned assembly drawings are provided as separate attachments:

- **Sphere Assembly Drawing 1:** Multi-view orthographic projections with isometric views at 1:5 scale.
- **Sphere Assembly Drawing 4:** Dimensioned assembly drawing (160.52 mm height, 52.88 mm radius, 43 mm base) with detail view at 1:2 scale.
- **Sphere Assembly Drawing 4 (Annotated):** Labelled part callouts identifying hose clamp insert, bolt/string insert, microphone plate, electrical storage, clips, hemispheres, and mesh area.

These drawings were generated from the Onshape CAD model (1050 Sound Lab 20) and are titled "L20 Sound Measuring Device."